ie UNIVERSITY | SCHOOL OF SCIENCE & TECHNOLOGY

BACHELOR IN DATA AND BUSINESS ANALYTICS

TIME SERIES ANALYSIS

Session 8 · ETS Taxonomy and Model Selection

Professor: Angel Mora Segura

08 October 2026 · 18:00–19:30 · Live in-person



COURSE SCHEDULE · 20 SESSIONS · 80 MINUTES EACH

Session by Session

1	02 Sep	Introduction · R setup · forecasting as a business problem ✓
2	03 Sep	Time plots, seasonal plots, EDA ✓
3	23 Sep	Lags, ACF, PACF ✓
4	24 Sep	Decomposition, STL, Box-Cox · Checkpoint 1 ✓
5	30 Sep	Benchmarks: Naive & Seasonal Naive ✓
6	01 Oct	Simple exponential smoothing ✓
7	07 Oct	Holt, damped trend, Holt-Winters ✓
8	08 Oct	ETS taxonomy + model selection **TODAY**
9	15 Oct	**Midterm (intermediate test) — 20% · Checkpoint 2**
10	19 Oct	AR / MA
11	22 Oct	ARMA + stationarity
12	28 Oct	ARMA + integration + diagnostics
13	29 Oct	ARIMA
14	12 Nov	Seasonal ARIMA (SARIMA)
15	18 Nov	Automatic ARIMA + model selection
16	19 Nov	Complex seasonality / Fourier
17	25 Nov	Integrated Forecasting Workshop · Checkpoint 3
18	02 Dec	**Group presentations — 30% · double session, part 1**
19	02 Dec	**Group presentations — 30% · double session, part 2**
20	11 Dec	**Final exam in R — 40%**

TODAY'S SESSION · LAST OF BLOCK 2

One Family of Models

The ETS taxonomy, what the error term adds, information criteria, automatic selection — and when to trust it.

FPP3 · Sections 8.4 – 8.7

TYPICAL CLASS STRUCTURE · 80 MINUTES

The Classroom Is the Laboratory

The same three blocks every session. The work on your case happens here, with me in the room.

~30 MIN

BLOCK 1

Concepts & R application

The methodology, its intuition and its assumptions — aimed at **what problem this technique solves** — then straight into its practical implementation in R.

You leave with code that runs.

~30 MIN

BLOCK 2

Business Case Lab

Teams apply or evaluate the methodology in the context of their own case. Finished code is not the requirement.

Output can be a plot, a methodological decision, a hypothesis, a data transformation, a fitted model, or a justified argument for why this technique does not apply.

~20 MIN

BLOCK 3

Business Case Clinic

Selected teams share their approach, findings or difficulties in about 5 minutes each.

The class discusses alternatives and gives feedback; I close the loop.

Block durations vary with the topic: technically demanding sessions take more time for explanation and R; later sessions shift progressively towards project development and feedback.

BY THE END OF TODAY YOU SHOULD BE ABLE TO

Session Objectives

1. Read and write **ETS(Error, Trend, Season)** notation, and place every method from Sessions 6 and 7 in the grid.

2. Explain what distinguishes a **model** from a method, and why the error type changes the intervals but not the point forecasts.

3. Use **AICc** to compare models, and state the conditions under which that comparison is valid.

4. Run **automatic ETS selection** in R and decide whether to accept its choice.

5. **Close Block 2 for your case**: the benchmark, the best ETS model and the evidence, ready for Checkpoint 2.

BLOCK 1 · 30 MINUTES

Concept + R Application

From a collection of methods to a family of models.

IN CLASS: THE CORE IDEAS · ANNEX + QUARTO: EQUATIONS, R DETAILS AND THE REST

The ETS Grid

Three choices of trend, three choices of season. Every smoothing method you have met sits in one cell. The error type — additive or multiplicative — doubles the grid into models.

	SEASON: NONE (N)	SEASON: ADDITIVE (A)	SEASON: MULTIPLICATIVE (M)
TREND: NONE (N)	`(N, N)` Simple exponential smoothing	`(N, A)` Seasonal, no trend	`(N, M)` Seasonal, no trend
TREND: ADDITIVE (A)	`(A, N)` Holt's linear trend	`(A, A)` Additive Holt-Winters	`(A, M)` Multiplicative Holt-Winters
TREND: DAMPED (Ad)	`(Ad, N)` Damped trend	`(Ad, A)` Damped additive Holt-Winters	`(Ad, M)` Damped multiplicative Holt-Winters

Dark cells: the methods covered in Sessions 6 and 7. The cell gives Trend and Season; the full name adds the Error in front, e.g. ETS(M, A, M).

Reading ETS(Error, Trend, Season)

Three letters, always in this order. Once you can read them, a model report tells you what the model assumes about the series in a single glance.

ETS(A, N, N)

Simple exponential smoothing with additive errors. A level and nothing else.

ETS(A, Ad, N)

Damped trend, no season. The standard choice for trending annual data.

ETS(M, A, M)

Multiplicative Holt-Winters with multiplicative errors. The seasonal swing and the noise both grow with the level.

ETS(A, A, A)

Additive Holt-Winters. Constant-width season, constant-size noise.

ETS(M, A, M) is the model behind yesterday's winning Holt-Winters on Happy Cow. It now has a name, and a likelihood.

Comparing Models: AICc

A more complex model always fits the training data at least as well. An information criterion rewards fit and **penalises every extra parameter**, so complexity has to pay for itself.

$$AIC = -2 \log L + 2k$$

L is the likelihood, k the number of estimated parameters. **Lower is better.** AICc adds a correction for small samples, and is the one R uses by default.

- Minimising AICc is asymptotically equivalent to minimising one-step cross-validated error — which is why it works as a shortcut for model choice.

The rules for a valid comparison

- **Same data.** Every model must be fitted to exactly the same observations over the same period.
- **Same transformation.** You cannot compare the AICc of a model on `log(y)` with one on `y`.
- **Same family.** Compare ETS models with each other. To compare ETS with ARIMA in Block 3, use test-set accuracy instead.
- The absolute value means nothing. Only the differences between models matter.

AICc ranks models on the training data. The test set still decides — a low AICc is not evidence that a model forecasts well.

R APPLICATION

Letting R Choose the Model

```
fit <- train |>
  model(
    snaive = SNAIVE(units),
    auto   = ETS(units)
  )
fit |> select(auto) |> report()
# example of the output format:
#> Series: units
#> Model: ETS(M,A,M)
#>   Smoothing parameters:
#>     alpha = ...  beta = ...
#>     gamma = ...
#>   AICc  ...
```

What `ETS(units)` does

- Fits every admissible combination of error, trend and season to the training data.
- Estimates the smoothing parameters and initial states of each by maximum likelihood.
- Returns the one with the lowest AICc, and its name tells you what it found.
- You can restrict the search: `ETS(units ~ trend(c("N","Ad")))` tries only no trend or a damped one.

A useful check: compare R's choice with the model you would have picked from your Block 1 diagnostics. If they disagree, find out why before accepting either.

When to Trust the Automatic Choice

Automatic selection is a good starting point and a poor final answer. It optimises one criterion on the training data; your recommendation has to hold on data it has not seen, for a decision it does not know about.

Trust it more when

The series is long, the patterns are clear, and the chosen model agrees with what your plots and ACF showed.

Trust it less when

The series is short (two or three seasonal cycles), contains a structural break, or the chosen model contradicts the diagnostics.

Always confirm

Put the selected model in the accuracy table with the benchmark. A low AICc is not evidence that it forecasts well.

Remember the purpose

AICc knows nothing about which error is expensive for the business. Choose among the top few models with your accuracy measure.

EXAMPLE HAPPY COW · EVERY MODEL FROM BLOCK 2 · TEST SET 2026

The Complete Block 2 Comparison

Model	RMSE	MAE	MASE
ETS(M,A,M)	**167**	**157**	**0.64**
ETS(M,Ad,M)	363	348	1.42
ETS(A,A,A)	430	376	1.54
Seasonal naive	**411**	**386**	**1.57**
Mean	1,628	1,343	5.48
Drift	2,376	1,814	7.40
Naive	2,447	1,896	7.74

This is what a Checkpoint 2 table looks like.

- Every model tried in Block 2, on the **same** test set, sorted by the chosen measure.
- The benchmark is highlighted, because it is the line everything is judged against.
- One model clears it by a wide margin: ETS(M,A,M), with MAE 157 against 386.
- The losing rows stay in the table. They are the evidence that the winner was not chosen by accident.

From Chosen Model to Forecast

Once the test set has confirmed the choice, refit the model on **all** the data and forecast the real future. The test set has done its job; throwing its information away now would only make the forecast worse.

```
# refit the confirmed model on everything
final <- sales |>
  model(ets = ETS(units ~ error("M") +
                  trend("A") + season("M")))
final |> gg_tsresiduals()
final |> forecast(h = 12) |>
  hilo(level = c(80, 95))
```

- **Check the residuals again** on the refitted model: uncorrelated, zero mean. The Session 5 checks apply unchanged.
- **Report the intervals** at the level your decision needs. For a stockout-averse client, the upper 80% or 95% bound is the planning number.
- **Keep the specification fixed** when you refit. Re-running automatic selection on the full data can choose a different model that was never tested.

The sequence for every model in this course: **choose on training → confirm on test → refit on all → forecast with intervals.**

BLOCK 2 · 30 MINUTES

Business Case Lab

Run automatic ETS on your case, compare it with your own choice, and assemble the Checkpoint 2 table.

WHAT TO EXPECT IN EACH CASE

Automatic ETS and Your Case

CASE 1 · RETAIL

Happy Cow Ice Cream

Expect a seasonal model with m = 7. Compare its choice of additive or multiplicative season with your Session 2 diagnosis.

Watch for: with seven months, the automatic search is working on very little data. Confirm the choice on the test set before relying on it.

CASE 2 · AIRLINES

Balearic Airlines

The chosen model matters less than its intervals: the protection level is a quantile. Compare the 80% intervals of the top two models.

Watch for: multiplicative error widens peak-day intervals — exactly the days where the protection decision is hardest.

CASE 3 · MANUFACTURING

Clean Air Limited

With 24 training months, restrict the search to damped or no trend. A year-ahead commitment does not need a trend that runs forever.

Watch for: if the break is in the training data, the automatic choice may be fitting COVID. Run it with and without those months.

CASE 4 · E-COMMERCE

Wayfair

Run `ETS()` on the Decor total and each of the four SKUs, and compare the models it picks. The differences are a finding in themselves.

Watch for: SKUs with zeros can only take additive models. Report which of the four that excludes from the multiplicative forms.

BLOCK 2 COMPLETE · SESSIONS 5 TO 8

Building the Forecasting Baseline: Done

SESSION 5

Set the bar

Benchmarks, train and test, residuals, accuracy measures, cross-validation.

SESSION 6

Weighted the past

Simple exponential smoothing and the meaning of α.

SESSION 7

Followed trend and season

Holt, the damped trend, Holt-Winters additive and multiplicative.

SESSION 8

Named the family

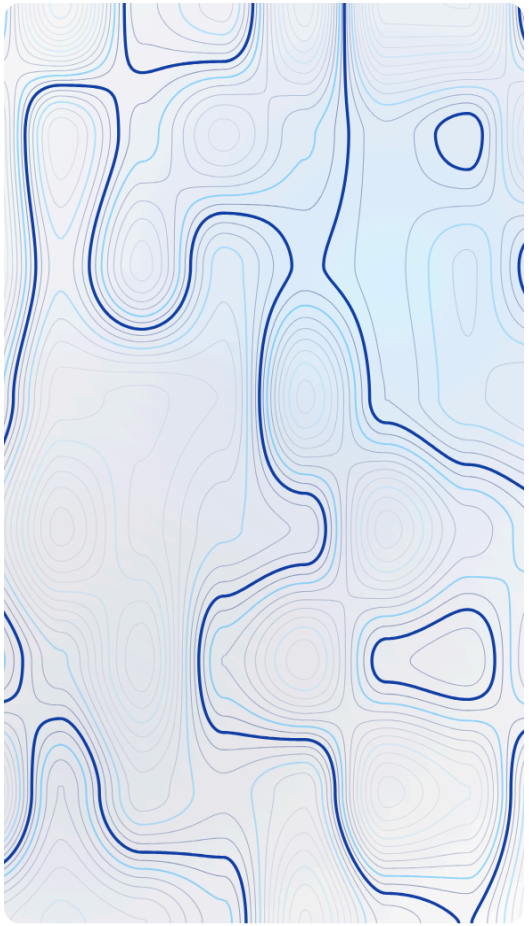
ETS notation, the error term, AICc, automatic selection and when to trust it.

BLOCK 3 · FROM SESSION 10

Statistical modelling

From Session 10: AR, MA and ARIMA, modelling the autocorrelation directly. Every ARIMA model will be compared against the ETS model you choose today.

Your ETS model is now the **second number to beat**. From Block 3, a new model has to improve on both the benchmark and the best ETS.



END OF SESSION 08

TIME SERIES ANALYSIS

ETS Taxonomy and Model Selection

NEXT · SESSION 9 · 15 OCTOBER

Midterm · intermediate test

Session 9 is the midterm only, covering Blocks 1 and 2. Block 3 — AR, MA and ARIMA — begins in Session 10.

The annex after this slide holds the reference material, including the R cheat sheet on slide 30.

Angel Mora Segura · IE University

08 October 2026